MapReduce-Based Taxi Fleet and Company Performance Analysis

Assignment One

Assignment name: MapReduce-Based Taxi Fleet and Company Performance Analysis

Course code: COSC2633/2637

Weighting: 25/100

Due date: 30/08 at 23:59

1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes:

- CLO 1: model and implement efficient big data solutions for various application areas using appropriately selected algorithms and data structures.

- CLO 2: analyse methods and algorithms, to compare and evaluate them with respect to time and space requirements and make appropriate design choices when solving real-world problems.
- CLO 3: motivate and explain trade-offs in big data processing technique design and analysis in written and oral form.
- CLO 4: explain the Big Data Fundamentals, including the evolution of Big Data, the characteristics of Big Data and the challenges introduced.
- CLO 6: apply the novel architectures and platforms introduced for Big data, i.e., Hadoop, MapReduce and Spark.

2. Overview of Assessment

Write MapReduce programs which give your chance to develop basic understanding of principles when solving complex problems on Hadoop execution platform.

Complete the given MapReduce programming tasks with Python (standard library only) and the methods taught in this course. Do not use any third-party Python packages.

You must apply the knowledge and skills acquired in this course to manage your HDFS directories.

3. Assessment Tasks

Follow the steps below to get started on this assessment task:

- **Assignment 1 Specification**
- **Trips.txt**
- **Taxis.txt**
- **initialization.txt**

- A1_GroupLeaderStudentID_GroupNumber_UGorPG_worklog.docx

4. Group Assessment

Groups must be formed by **2 or 3 students** who are enrolled at the same level (either all Undergraduate COSC2633 or all Postgraduate COSC2637). All members are expected to contribute equally. Unless a contribution dispute is reported (and supported by the work log), all group members will receive the same grade.

Individual work is permitted but not encouraged. Please note that no bonus marks will be awarded for completing the work individually.

- Navigate to **People** in Canvas, then select the **Groups**
- Choose an empty **A1 group** and join it along with your group members.
- **You are not permitted to create new groups. You are not permitted to rename the groups.** This restriction is in place because student-created groups are not linked to the grading system, which means they cannot be identified by the system during marking.
- Canvas groups are intended only for students completing the assignment in groups of two or three. Students completing the assignment individually do not need to join a Canvas group.

Please note that it is each student's responsibility to ensure they have successfully joined the Canvas group before commencing work on Assessment 1.

Group membership should be confirmed by individual students before work begins. The team leader's role is generally to verify the group composition before and after submission. However, this does not remove each team member's responsibility to ensure that they are correctly registered in the group throughout the assessment period.

Detailed requirements and instructions can be found in the assignment specification document.

5. Submission Instructions

Detailed requirements and instructions can be found in the assignment specification document.

Students who choose to complete the assessment collaboratively but fail to correctly form their group before the submission deadline will receive a 10% deduction (2.5 marks).

Final Submission Warning

Please verify your submission carefully before the deadline. Ensuring that the correct files have been submitted is **the responsibility of the entire group**, not just the group leader. All group members are expected to check that the submission is complete, correct, and submitted before the deadline.

Groups will be locked 72 hours before the deadline. You must finalise and verify your group membership **before submission** and only submit after groups are locked. **Please do not submit your assignment until the groups have been finalised and locked.**

- You must submit 3 files: **the zip file, the doc file, and the pdf file** for Turnitin plagiarism check.
- Failure to submit the required pdf or doc file separately from the zip file will result in a 10% penalty (2.5 marks).
- Submissions that do not include the required pdf file will incur a 10% penalty (2.5 marks).
- Submissions that do not include the required doc work log document will incur a 20% penalty (5 marks).
- Submissions where the pdf **fails the Turnitin plagiarism check**, **will not be graded**.
- Please verify your submission carefully before the deadline. If you submit the **wrong file**, a **corrupted zip file**, or **a submission that does not comply with the required file structure**, any resubmission made after the deadline will be subject to the standard late submission penalty.

- Late submission penalties are applied automatically by the system based on the assessment due date and the time of submission. **The penalty is determined by when you submit your work, not when you complete it.** Therefore, even if you completed the assessment before the due date, a late submission would still incur the applicable late penalty.
- Work must run on the **RMIT-provided EMR.**
- **Please note that markers will not modify or adjust submitted code during marking.**

Please carefully review the detailed submission requirements in the assignment specification. The marking rubric clearly states the mark deductions for failing to meet these requirements.

6. Assessment Criteria

Assessment for this MapReduce assignment will be based on two primary criteria:

1. The correctness of the program's output: the program must produce accurate and complete results as specified in the task requirements.
2. The correctness of the implementation: your code should demonstrate a sound understanding of MapReduce principles, including
    - Efficient handling of large-scale (big) data.
    - Your implementation should leverage MapReduce's parallel processing model by correctly utilising multiple mappers and reducers to maximise scalability and performance.
    - Correct and efficient use of MapReduce components: combiner, partitioner, and data processing pipeline.

Marks will be awarded based on the quality of your design and implementation, rather than solely on producing the correct output. Solutions should demonstrate a sound understanding of MapReduce principles and make effective use of the framework's parallel processing capabilities.The marking guide is included in assignment specification.

Criteria	Ratings	Pts
Task 1 See detailed marking guide in A1 specification 1. -2.5 marks: The program did not implement in-mapper combining with state preserved across lines. 2. -0.5 mark: Your output didn't follow Output Format Requirements specified in the requirements. 3. -1 mark: Your bash script did not work. 4. -1 mark each: Incorrect aggregation (total/min/max/avg).		5 pts
Task 2 See detailed marking guide in A1 specification Mapper: 3 marks Bash script: 3 marks + executes the MapReduce jobs in the correct sequence, + passes parameters correctly, + manages HDFS input/output directories, + terminates when the algorithm converges or reaches the maximum number of iterations, + produces the expected std output Reducer: 4 marks		10 pts
Task 3 (Postgraduate COSC2637) and Task 4 (Undergraduate COSC2633) See detailed marking guide in A1 specification ========== Task 3 (Postgraduate COSC2637) Join operation: 3 marks Aggregation operation: 4 marks Sorting: 3 marks ==========		10 pts

Criteria	Ratings	Pts
Task 4 (Undergraduate COSC2633) Job design and automation: 3 marks Mean: 3 marks Std. Deviation: 4 marks		
Format Requirements Failure to follow the format requirements incurs up to 10 marks penalty. Please read the spec carefully. 1. -5 marks: Work log MS Word file missing. 2. -2.5 marks: Your ZIP file must not contain any sub-folders. 3. -1 mark: Your zip/doc/pdf file is incorrectly named. 4. -2.5 marks: HDFS input OR output dir is incorrectly specified. 5. -2.5 mark: Bash script is incorrectly named. 6. -2.5 marks: The PDF file for Turnitin plagiarism check must be submitted separately from the ZIP file. If the PDF is included only inside the ZIP or is missing entirely, a 10% penalty will apply.		0 pts
Functional Requirements Failure to follow the functional requirements incurs up to 5 marks penalty. Please read the spec carefully. 1. -2.5 marks: The code is not well written using a good coding style. 2. -2.5 marks: The codes and scripts did not come with concise and clear comments to explain the logical flow of the program.		0 pts
		Total Points: 25